MATH 2211, SPRING 2026: HOMEWORK 10

(1) Suppose that $T : V \to V$ is a linear transformation and that $\mathcal{B}_1$ and $\mathcal{B}_2$ are bases for V. Show that $[T]_{\mathcal{B}_1}$ and $[T]_{\mathcal{B}_2}$ have the same minimal polynomial.

(2) What are the minimal and characteristic polynomials of the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

?

(3) Suppose that $A \in M_{n\times n}(F)$ is such that $\min_A(x)$ is a multiple of $(x-\lambda)^2$ for some $\lambda \in F$. Show that A cannot be diagonalizable.

Definition 1. If $A \in M_{n\times n}(\mathbb{C})$ and $\lambda \in \mathbb{C}$, then the **generalized λ-eigenspace for A** is

$$E_\lambda^{\text{gen}} = \{v \in \mathbb{C}^n : (A - \lambda I_n)^m v = 0 \text{ for some } m \geq 1\}.$$

A non-zero vector in E_λ^{gen} is called a **generalized eigenvector** with eigenvalue λ.

(4) Show that the following are equivalent for an eigenvalue $\lambda \in \mathbb{C}$ for A:
  (a) $E_\lambda^{\text{gen}} = E_\lambda$.
  (b) The minimal polynomial $\min_A(x)$ is a multiple of $x - \lambda$ but not of $(x-\lambda)^2$.

Assume the following result:

Theorem 1. *F^n admits a basis consisting of* generalized *eigenvectors for A.*

(5) Using the above theorem, show that $\dim E_\lambda^{\text{gen}} = m(\lambda)$, where $m(\lambda)$ is the algebraic multiplicity, and conclude that A is diagonalizable whenever $\min_A(x)$ has no repeated linear factors.

(6) Use least squares approximation to find the cubic function $f(x) = a_3x^3 + a_2x^2 + a_1x + a_0 \in \text{Poly}(\mathbb{R})$ that best minimizes the quantity

$$f(-2)^2 + (f(-1)+1)^2 + f(0)^2 + (f(1)-2)^2 + (f(2)-1)^2.$$

(7) Consider

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \in M_{3\times 3}(\mathbb{R})$$

Use the Gram-Schmidt process to write this as a prodcut $A = QR$, where Q is orthonormal and where R is upper triangular.

(8) Suppose a bacterial population $P(t)$ (in thousands) is observed at time t (in hours). The growth follows an exponential model:

$$P(t) = Ae^{kt}$$

where A is the initial population and k is the growth rate. Given the following data, find the best fit for A and k using the method of least squares.

Time t (hr)	1	2	3	4
Population P	2.1	4.5	9.2	18.8

Hint: Linearize the problem by taking logarithms!